

Configure the Standard Access Control List using cisco packet tracer and verify the configuration



Dr. G. Omprakash

Assistant Professor, ECE, KLEF



Access Control lists

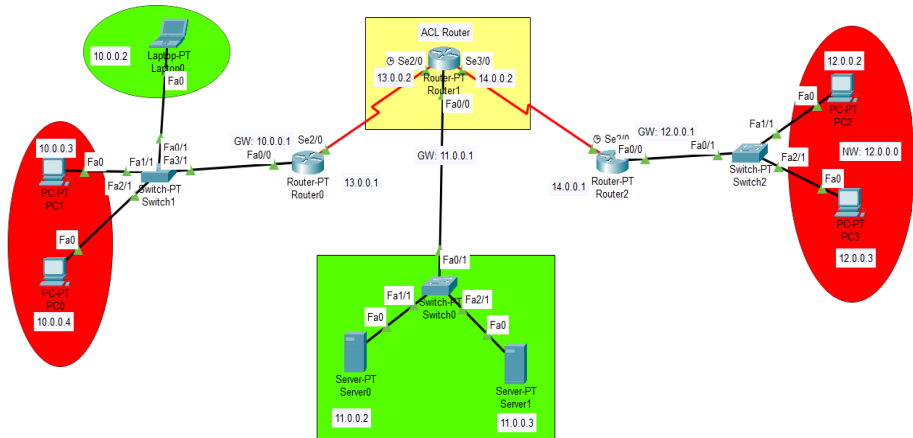
- Access Control Lists (ACLs) is a set of rules defined for controlling network traffic and reducing network attacks.
- Primarily used for security purposes, filtering traffic, and defining access permissions in routers
- **Types of ACL**
 - **Standard Access Control List**
 - Standard ACL filters traffic based **only on source IP addresses**
 - Range of Standard ACL Numbers: 1–99 and 1300–1999
 - Should be applied **near the destination**
 - **Extended Access Control List**
 - An Extended ACL filters traffic based on several criteria: source and destination IP addresses, protocols (TCP, UDP, ICMP), and port numbers.
 - Range of Extended ACL Numbers: 100–199 and 2000–2699
 - Should be applied as close to the source as possible



Standard Access Control List



Standard Access Control List





Devices and Connections

Aim: The access to the servers should be given only to the laptop.

Other PCs can communicate to each other using the network

- Select three routers (Router-PT) and place them on the workspace
- Select three switches (Switch-PT) and place them on the workspace
- Select four PCs and laptop and place them on the workspace
- Select two servers (Server-PT) and place them on the workspace
- Connections for left-side network
 - Use a Copper straight-through and connect
 - FastEthernet of PC0 $\Rightarrow$ FastEthernet2/1 of the switch1
 - FastEthernet of PC1 $\Rightarrow$ FastEthernet1/1 of the switch1
 - FastEthernet0/1 of the switch1 $\Rightarrow$ FastEthernet of the laptop
 - FastEthernet3/1 of the switch1 $\Rightarrow$ FastEthernet0/0 of the Router0



Connections

- Connections for Middle part of the network
 - Use a Copper straight-through and connect
 - FastEthernet0/0 of Router1 $\implies$ FastEthernet0/1 of the Switch0
 - FastEthernet of Server0 $\implies$ FastEthernet1/1 of the Switch0
 - FastEthernet of Server1 $\implies$ FastEthernet2/1 of the Switch0
 - Serial2/0 of the Router1 $\implies$ Serial2/0 of the Router0
 - Serial3/0 of the Router1 $\implies$ Serial2/0 of the Router2
- Connections for the right-side of the network
 - Use a Copper straight-through and connect
 - FastEthernet of PC3 $\implies$ FastEthernet2/1 of the switch2
 - FastEthernet of PC2 $\implies$ FastEthernet1/1 of the switch2
 - FastEthernet0/1 of the switch2 $\implies$ FastEthernet0/0 of the Router0



Configuring Gateway of Router0 (Left)

Router0

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0**
- FastEthernet1/0
- Serial2/0
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0030.F20C.3BBA

IP Configuration

IPv4 Address 10.0.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ip address 10.0.0.1 255.0.0.0
Router(config-if)#
```



Serial port Configuration of Router0 (Left)

Router0

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0
- FastEthernet1/0
- Serial2/0**
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

Serial2/0

Port Status

Duplex ☒ Full Duplex ☐ On

Clock Rate 1200

IP Configuration

IPv4 Address 13.0.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config-if)#  
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up  
ip address 10.0.0.1 255.0.0.0  
Router(config-if)#ip address 10.0.0.1 255.0.0.0  
Router(config-if)#  
Router(config-if)#exit  
Router(config)#interface Serial2/0  
Router(config-if)#no shutdown  
Router(config-if)#ip address 13.0.0.1 255.0.0.0  
Router(config-if)#
```



Configuring Gateway of Router1 (Middle)

Router1

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0**
- FastEthernet1/0
- Serial2/0
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0000.0CE3.E858

IP Configuration

IPv4 Address 11.0.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ip address 11.0.0.1 255.0.0.0
Router(config-if)#
```



Serial port (S2) Configuration of Router1 (Middle)

Router1

Physical Config CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0
- FastEthernet1/0
- Serial2/0**
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

Serial2/0

Port Status

Duplex ☒ Full Duplex ☐ On

Clock Rate 1200

IP Configuration

IPv4 Address 13.0.0.2

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

```
ip address 11.0.0.1 255.0.0.0
Router(config-if)#ip address 11.0.0.1 255.0.0.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
ip address 13.0.0.2 255.0.0.0
Router(config-if)#
```



Serial port (S3) Configuration of Router1 (Middle)

Router1

Physical Config CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0
- FastEthernet1/0
- Serial2/0
- Serial3/0**
- FastEthernet4/0
- FastEthernet5/0

Serial3/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 1200

IP Configuration

IPv4 Address 14.0.0.2

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config-if)#  
%LINK-5-CHANGED: Interface Serial2/0, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up  
ip address 13.0.0.2 255.0.0.0  
Router(config-if)#ip address 13.0.0.2 255.0.0.0  
Router(config-if)#  
Router(config)#exit  
Router(config)#interface Serial3/0  
Router(config-if)#no shutdown  
Router(config-if)#ip address 14.0.0.2 255.0.0.0  
Router(config-if)#
```



Configuring Gateway of Router2 (Right)

Router2

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0**
- FastEthernet1/0
- Serial2/0
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 00D0.BAB0.50A7

IP Configuration

IPv4 Address 12.0.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 12.0.0.1 255.0.0.0
Router(config-if)#ip address 12.0.0.1 255.0.0.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
```



Serial port (S2) Configuration of Router2 (Right)

Router2

Physical Config CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

INTERFACE

- FastEthernet0/0
- FastEthernet1/0
- Serial2/0**
- Serial3/0
- FastEthernet4/0
- FastEthernet5/0

Serial2/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 1200

IP Configuration

IPv4 Address 14.0.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

Equivalent IOS Commands

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
ip address 14.0.0.1 255.0.0.0
Router(config-if)#
```



Assign IP Address

Click on PC/Server → Desktop Tab → IP Configuration

- Laptop0: IPv4: 10.0.0.2; SubMask:255.0.0.0; Gateway: 10.0.0.1
- PC0: IPv4: 10.0.0.4; SubMask:255.0.0.0; Gateway: 10.0.0.1
- PC1: IPv4: 10.0.0.3; Sub Mask:255.0.0.0; Gateway: 10.0.0.1
- PC2: IPv4: 12.0.0.2; SubMask:255.0.0.0; Gateway: 12.0.0.1
- PC3: IPv4: 12.0.0.3; Sub Mask:255.0.0.0; Gateway: 12.0.0.1
- Server0: IPv4: 11.0.0.2; Sub Mask:255.0.0.0; Gateway: 11.0.0.1
- Server: IPv4: 11.0.0.3; Sub Mask:255.0.0.0; Gateway: 11.0.0.1



Routing Configuration (RIP)

• Router 0

- Router(config)#router rip
- Router(config-router)#network 10.0.0.0
- Router(config-router)#network 13.0.0.0
- Router(config-router)#exit

• Router 1

- Router(config)#router rip
- Router(config-router)#network 13.0.0.0
- Router(config-router)#network 11.0.0.0
- Router(config-router)#network 14.0.0.0
- Router(config-router)#exit

• Router 2

- Router(config)#router rip
- Router(config-router)#network 14.0.0.0
- Router(config-router)#network 12.0.0.0
- Router(config-router)#exit



Check the Connections before ACL Configuration

- Ping the PCs to check connection. Every PC is connected to every other PC
- Inspect the Routing table
- "sh ip route rip" in the CLI of the router

Laptop0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 11.0.0.2

Pinging 11.0.0.2 with 32 bytes of data:

Reply from 11.0.0.2: bytes=32 time=20ms TTL=126
Reply from 11.0.0.2: bytes=32 time=1ms TTL=126
Reply from 11.0.0.2: bytes=32 time=25ms TTL=126
Reply from 11.0.0.2: bytes=32 time=1ms TTL=126

Ping statistics for 11.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 25ms, Average = 11ms

C:\>ping 12.0.0.2

Pinging 12.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 12.0.0.2: bytes=32 time=2ms TTL=125
Reply from 12.0.0.2: bytes=32 time=31ms TTL=125
Reply from 12.0.0.2: bytes=32 time=17ms TTL=125

Ping statistics for 12.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 31ms, Average = 16ms

C:\>|
```



Configuring Standard ACL at Router 1

- Router(config)#access-list 10 deny host 10.0.0.3
- Router(config)#access-list 10 deny host 10.0.0.4
- Router(config)#access-list 10 deny 12.0.0.0 0.0.0.255
- Router(config)#access-list 10 permit any
- Router(config)#int f0/0
- Router(config-if)#ip access-group 10 out
- Router(config-if)#exit
- Use "out" when you want to filter traffic leaving the router from the interface f0/0.
- "access-list 10 deny 12.0.0.0 0.0.0.255" It blocks all traffic originating from the 12.0.0.0/24 subnet.
- 0.0.0.255 means the first three octets (12.0.0) must match exactly, but the last octet can be any value (0-255).



Checking Standard ACL

To check whether the standard access list is working properly or not, we ping the Server(11.0.0.2) from the PC0 (10.0.0.4) which had been blocked and we get the following result.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 11.0.0.2

Pinging 11.0.0.2 with 32 bytes of data:

Reply from 13.0.0.2: Destination host unreachable.
Reply from 13.0.0.2: Destination host unreachable.
Reply from 13.0.0.2: Destination host unreachable.
Reply from 13.0.0.2: Destination host unreachable.
```



Conclusion

Implementation of ACLs ensures that only authorized traffic is allowed, protecting network resources from unauthorized access and potential threats